

Functional Test Defect Report

Project: Property Management System

Field	Details
Group Name	Commented Code
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Course / Institution	Software Verification, Validation, Security / Fatih Sultan Mehmet Vakıf Üniversitesi
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System Overview

This report documents the functional test suite for the **Software Verification, Validation, Security** project. The system architecture utilizes a React frontend, an Express backend, and a PostgreSQL database. Security is maintained via JWT-based (Bearer Token) authentication.

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Defect Resolution Report: Authentication & Security

Project: Real Estate Platform

Testing Cycle: Phase 1 (Auth & Session Management)

Status: All Resolved

Defect ID	Associated Test Case	Summary of Issue	Resolution / Fix Applied	Status
BUG-REG-011	TC-REG-011	XSS Validation Gap: Payload handling couldn't be verified because the <code>fullname</code> was not rendered anywhere in the UI.	Updated the User Profile and Navbar components to display the <code>fullname</code> . Verified that React automatically escapes the script payload.	FIXED
BUG-REG-012	TC-REG-012	Security Vulnerability: Lack of rate limiting allowed for unlimited rapid account creation (Brute-force/DoS risk).	Implemented <code>express-rate-limit</code> on the <code>/api/auth/register</code> endpoint. Limited to 5 attempts per 15-minute window per IP.	FIXED
BUG-LOG-011	TC-LOG-011	UI Interaction Failure: The logout test failed because the button lacked a unique identifier for the test runner.	Assigned <code>id="logout-btn"</code> to the component. Verified that clicking it clears <code>localStorage</code> and resets the Auth context.	FIXED
BUG-LOG-013	TC-LOG-013	Test Script Error: Backend was rejecting missing headers, but the test code was not asserting the correct 401 status.	Refactored the Playwright test to correctly expect and assert a <code>401 Unauthorized</code> response when headers are null.	FIXED

BUG-LOG-014	TC-LOG-014	Middleware Edge Case: Improperly formatted headers (missing "Bearer" prefix) were not being caught correctly by the test script.	Updated the backend middleware to strictly validate the "Bearer " prefix and updated the test suite to verify this rejection.	FIXED
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Defect Resolution Report: Property Management (MP & EDIT)

Project: Real Estate Platform

Testing Cycle: Phase 2 (Make, Read, Update, Delete & Validation)

Status: 3 Fixed / 2 Open (Technical Blockers)

Defect ID	Test Case	Summary of Issue	Resolution / Current Status	Status
BUG-E DIT-00 1	TC-FU NC-EDI T-001	Logic Error: Owner could not update properties due to incorrect ID parsing in the controller.	Refactored controller to correctly extract <code>:id</code> from <code>req.params</code> . Verified successful DB update.	FIXED
BUG-E DIT-00 2	TC-FU NC-EDI T-002	Security Flaw: Ownership check was missing; unauthorized users could potentially hit update endpoints.	Added an authorization check: <code>if (property.owner_id !== req.user.id) return 403.</code>	FIXED
BUG-E DIT-00 7	TC-FU NC-EDI T-007	Data Integrity: Database accepted negative prices or non-numeric strings for property listings.	Integrated Zod schema: <code>z.coerce.number().positive()</code> . Invalid inputs now return 400 Bad Request.	FIXED
BUG- MP-00 5	TC-FU NC-MP -005	Script Error: Playwright <code>apiRequestContext.post</code> failed with a Stream error during multi-image upload simulation.	Investigation: Likely caused by malformed <code>multipart/form-data</code> in the test script. Needs <code>fs.readSync</code> fix.	OPEN
BUG- MP-00 7	TC-FU NC-MP -007	Timeout Error: AI generation test timed out waiting for a response when no images were sent.	Investigation: Backend might be hanging (no <code>res.send</code>) instead of returning an error when <code>req.files</code> is empty.	OPEN

Defect Resolution Report: User Interactions & Dashboards

Project: Real Estate Platform

Testing Cycle: Phase 3 (Business Logic & Personalization)

Status: All Resolved

Defect ID	Test Case	Summary of Issue	Resolution / Fix Applied	Status
BUG-F AV-006	TC-FUNC-F AV-006	Data Integrity: Attempting to favorite a non-existent ID caused a DB Foreign Key error or a silent failure.	Added a guard clause in the controller to check if the property exists before inserting into the favorites table. Returns 404 Not Found .	FIXED
BUG-D ASH-001	TC-FUNC-M YLIST-001 TC-FUNC-M YSAVED-001	Auth Mismatch: Users were receiving 401 Unauthorized on their own dashboards even with valid tokens.	Root Cause: Email casing inconsistency (e.g., User@ex.com vs user@ex.com). Implemented .toLowerCase() and .trim() normalization.	FIXED
BUG-O FF-009	TC-FUNC-O FF-009	Negative Path: Offers could be submitted for phantom property IDs.	Added validation to ensure the target property_id is valid and active before accepting the offer.	FIXED
BUG-O FF-010	TC-FUNC-O FF-010	Logic Flaw: Sellers were able to "offer" on their own listings, inflating engagement metrics.	Added a comparison: if (listing.owner_id === buyer.id) return 403.	FIXED

BUG-O FF-011	TC-FUNC-O FF-011	Spam Control: Buyers could flood a seller with multiple pending offers for the same house.	Implemented a check for existing 'PENDING' offers from the same user on that ID. Returns 409 Conflict .	FIXED
BUG-O FF-013	TC-FUNC-O FF-013	State Integrity: Accepted/Rejected offers could be modified after the fact.	Added a "Final State" guard. Status can only be changed if the current status is strictly 'PENDING'.	FIXED